

Question

This is the Python scipy/numpy port of original MATLAB software. MATLAB was thoroughly verified with the desired results. I feel that the opening of the manual multitone.md contains a wrong assumption of mathematical nature: can You spot what it is?

CoT

Analyzing the Core Issue

I'm now zeroing in on the user manual's introduction to mult. My focus is entirely on pinpointing any flaws or misguided assumptions in how it presents the theoretical underpinnings. I'm carefully reviewing the initial statements to find a potential misconception or incorrect assumption, to analyze the foundation's validity.

Examining the Prime Tones

I'm now closely examining the use of prime numbers for audio test signals. I'm focusing on the claim that prime frequencies avoid harmonic relationships, leading to more distributed intermodulation products. I've taken note of two relevant text segments that emphasize this distribution benefit, and I'm focused on the relationship between harmonicity and intermodulation.

Investigating IMD Distribution

My current focus is the claim that prime frequencies distribute intermodulation (IMD) products evenly. I've now mathematically verified that distinct primes are not harmonically related, but I've realized that the resulting IMD products *are* still integers, so the "distribution" isn't as free as initially implied. I'm reassessing the manual's statement: "Primes are, by definition, not multiples of each other, so intermodulation products are distributed across the spectrum...". The claim seems incomplete, since integer products will form integer frequencies.

Revising IMD Product Claims

I'm now fully immersed in dissecting the "stacking" argument regarding IMD products. I've realized prime frequencies, although not harmonically related, produce IMD products that still land on integer bins. This isn't a "spread" in the way I initially thought. I am now focused on evaluating the integer results of the differences and sums. For any two prime frequencies in the test signal, the IMD will *always* fall on even integers.

Clarifying Stacking Behavior

I'm now clarifying the stacking behavior of integer IMD products on the 1 Hz grid. I am focused on whether IMD products stack on test tones. Second-order IMD will never fall on test tones because our primes are all odd. Third-order IMD can stack on test tones. The false assumption is the *reason* IMD doesn't stack. The fact that primes are...

Refining Harmonic Conflation

I'm now zeroing in on the text's conflation of harmonics with intermodulation products in the user manual. It mistakenly claims prime frequencies' "distribution" is superior because they aren't multiples, which fundamentally misrepresents IMD. I've realized, based on the text segments, that the frequencies *are* integer multiples of 1 Hz, which creates IMD products that fall on the integer grid. Odd primes create even second-order IMD products that fall on empty bins, and 3rd-order products may stack on test tones.

Defining the Ideal Case

I now understand that the manual's error lies in the flawed assumption about IMD distribution. The statement conflates "non-multiples" with a truly non-harmonic scenario. I now see the IMD products formed by primes *do* stack on integer bins. The ideal, truly non-harmonically related setup, would need logarithmically spaced, non-integer frequencies or even irrational numbers to avoid any stacking behavior. I am focused on refining the exact language describing the ideal case.

Synthesizing Prime Benefits

I'm now synthesizing the benefits of using prime frequencies. I understand the manual's flaw: prime numbers *do* produce stacked integer IMD products. The real benefit stems from odd/even arithmetic. Since primes are odd, 2nd-order distortion products fall on even bins and 3rd-order products may fall on test tones. This leads to reduced collisions compared to other integer spacing. My next step will be clarifying the math and the specific quote, linking all relevant sections.

Articulating Prime Spacing

I'm now articulating the specific benefits of prime number spacing in test tones. Second-order IMD products, being sums and differences of odd primes, are even and won't overlap test tones. Third-order products, although odd, have a reduced probability of collision due to the non-uniform distribution of primes. I'm focusing on contrasting this with standard harmonic grids.

Review by Gemini

You have a keen eye. The mathematical misconception lies in **Section 2.1: Prime-Based Frequency Selection**, specifically in this paragraph:

"If two tones at frequencies f_1 and f_2 are harmonically related (e.g., $f_2 = 2 \cdot f_1$), then their intermodulation products (sum and difference frequencies) fall back on the same harmonic grid, reinforcing each other and creating localized peaks. Primes are, by definition, not multiples of each other, so intermodulation products are distributed across the spectrum..."

The Misconception

The manual assumes that because primes are not multiples of each other, their Intermodulation Distortion (IMD) products will not "stack" on a grid and will instead be smoothly "distributed across the spectrum."

Mathematically, this is false. **All prime numbers are integers.** Therefore, any sum or difference of prime frequencies will always result in another integer.

Because of this, the IMD products absolutely *do* fall on a strict harmonic grid (a 1 Hz spacing grid), and they *do* stack precisely on top of one another on those integer bins. They are not arbitrarily distributed across the spectrum; they are strictly quantized to integer frequencies.

The Actual Mathematical Reason Primes are Used

The real reason prime frequencies are brilliant for multitone testing isn't because they avoid stacking entirely, but because **they avoid stacking on the test tones themselves.** This relies on basic integer factorization and even/odd parity, which the manual completely misses:

1. **Harmonic Distortion (HD) Avoidance:** By definition, a prime number has no divisors other than 1 and itself. Therefore, the harmonics of a prime ($2f, 3f, 4f...$) are

mathematically **composite numbers**. Because all other test tones are primes, a harmonic of one test tone can *never* land on another test tone.

2. **2nd-Order IMD Avoidance (The Even/Odd Trick):** In the audible range (above 2 Hz), **all prime numbers are odd**. Second-order intermodulation products are the sums and differences of two frequencies ($f_1 + f_2$ and $f_1 - f_2$). Mathematically, $Odd + Odd = Even$, and $Odd - Odd = Even$. Therefore, 100% of the 2nd-order IMD products will fall on **even integer bins**. Since all of your test tones are odd, 2nd-order distortion will *never* mask or overlap a test tone.
3. **3rd-Order IMD Mitigation:** Third-order products (e.g., $2f_1 - f_2$ or $f_1 + f_2 - f_3$) involve three terms. $Odd + Odd - Odd = Odd$. Because the result is odd, 3rd-order IMD *can* theoretically land on another prime number (test tone). However, because prime numbers become increasingly pseudo-randomly distributed as you move up the number line, the statistical likelihood of these odd-numbered IMD products landing exactly on *your specific prime tones* is vastly minimized compared to using linearly or logarithmically spaced integer tones.

Summary

The manual conflates “not being multiples of each other” with “not producing overlapping sums and differences.” The IMD products of primes still stack heavily on a 1 Hz grid—they just neatly stack on the **even and composite** bins, leaving the prime bins (where your test signal lives) beautifully clean for analysis!